

1. Cleaning Student Data Using NumPy and Pandas

Suppose the student dataset contains **Name, Marks, and Grade**, with some missing marks and inconsistent grade formats.

Simple Steps

Step 1: Load the dataset

```
import pandas as pd
import numpy as np

df = pd.read_csv("students.csv")
```

Step 2: Check missing values

```
print(df.isnull().sum())
```

Step 3: Fill missing marks

Use the average marks to replace missing values.

```
df["Marks"] = df["Marks"].fillna(
    df["Marks"].mean()
)
```

Step 4: Standardize grades

Convert grades to uppercase and remove unnecessary spaces.

```
df["Grade"] = (
    df["Grade"]
    .str.strip()
    .str.upper()
)
```

For example:

```
a → A
A → A
b → B
```

Step 5: Validate the cleaned data

```
print(df.isnull().sum())
```

```
print(df["Grade"].unique())
```

Conclusion

The process is:

Load Data



Check Missing Values



Fill Missing Marks



Remove Extra Spaces



Convert Grades to Uppercase



Validate Data

This produces a clean and consistent student dataset.

2. Total and Average Sales by Product Category

Pandas provides **groupby()** and **agg()** functions for this task.

Suppose the dataset contains:

Product	Category	Sales
Laptop	Electronics	50000
Mobile	Electronics	30000
Shirt	Clothing	2000
Jeans	Clothing	3000

Step 1: Load the data

```
import pandas as pd
```

```
df = pd.read_csv("sales.csv")
```

Step 2: Group by category

```
grouped = df.groupby("Category")
```

Step 3: Calculate total and average sales

```
result = grouped["Sales"].agg(  
    Total_Sales="sum",  
    Average_Sales="mean"  
)
```

Step 4: Display the result

```
print(result)
```

Output

	Total_Sales	Average_Sales
Category		
Clothing	5000	2500
Electronics	80000	40000

Simple Explanation

- **groupby("Category")** → Groups sales according to product category.
- **sum** → Calculates total sales for each category.
- **mean** → Calculates average sales for each category.
- **agg()** → Performs multiple calculations at the same time.

Conclusion

The basic approach is:

Load Sales Data

↓

Group by Category

↓

Calculate SUM

↓

Calculate AVERAGE

↓

Display Summary

This makes it easy to compare the sales performance of different product categories.

